

Geometry: Solving Kites Using Properties

Practice Worksheet • numberbender.com



Name: _____

Date: _____

Score: / 8

DIRECTIONS

Use kite properties to solve for x and find missing measures. $ABCD$ is a kite where $AB=AD$ and $CB=CD$, with AC as the main diagonal.

- 1 Kite — Consecutive sides equal. Solve for x :

$$AB = 3x + 2, \quad AD = 5x - 8$$

Answer: _____

- 2 Kite — Other pair of sides equal. Solve for x :

$$CB = 4x - 3, \quad CD = 2x + 7$$

Answer: _____

- 3 Kite — Opposite angles equal. Find x and $m\angle B$:

$$\angle B = (4x - 5)^\circ, \quad \angle D = (2x + 15)^\circ$$

Answer: _____

- 4 Kite — Diagonal bisects vertex angle $\angle A$. Find x :

$$\angle BAC = (2x + 8)^\circ, \quad \angle DAC = (4x - 12)^\circ$$

Answer: _____

- 5 Kite — Main diagonal bisects cross diagonal. Find x :

$$BE = 3x - 1, \quad DE = 4x - 7$$

Answer: _____

- 6 Kite — Vertex angle bisected at C . Find x :

$$\angle BCA = (x + 12)^\circ, \quad \angle DCA = (3x - 16)^\circ$$

Answer: _____

- 7 Kite — Find x and $m\angle B$. Use sum of angles = 360° :

$$\angle B = \angle D = (3x + 5)^\circ, \quad \angle A = 80^\circ, \quad \angle C = 60^\circ$$

Answer: _____

- 8 Kite — Find x and the perimeter:

$$AB = AD = 4x - 1, \quad CB = CD = 2x + 3, \quad P = 40$$

Answer: _____



Answer Key & Solutions

Geometry: Solving Kites Using Properties • Numberbender



TEACHER NOTES

Kite: 2 pairs consecutive sides equal. $\angle B = \angle D$ (opp. angles between unequal sides). Main diagonal (AC) bisects vertex angles and bisects BD. Diagonals are perpendicular.

1 Kite — Consecutive sides equal. Solve for x:

$$AB = 3x + 2, \quad AD = 5x - 8$$

$$= x = 5, \quad AB = AD = 17$$

Consecutive sides equal: $3x+2=5x-8 \rightarrow 10=2x \rightarrow x=5$.

2 Kite — Other pair of sides equal. Solve for x:

$$CB = 4x - 3, \quad CD = 2x + 7$$

$$= x = 5, \quad CB = CD = 17$$

$CB=CD$: $4x-3=2x+7 \rightarrow 2x=10 \rightarrow x=5$.

3 Kite — Opposite angles equal. Find x and $m\angle B$:

$$\angle B = (4x - 5)^\circ, \quad \angle D = (2x + 15)^\circ$$

$$= x = 10, \quad m\angle B = m\angle D = 35^\circ$$

$\angle B = \angle D$ (angles between unequal sides): $4x-5=2x+15 \rightarrow 2x=20 \rightarrow x=10$.

4 Kite — Diagonal bisects vertex angle $\angle A$. Find x:

$$\angle BAC = (2x + 8)^\circ, \quad \angle DAC = (4x - 12)^\circ$$

$$= x = 10, \quad m\angle BAC = m\angle DAC = 28^\circ$$

Main diagonal bisects $\angle A$: $\angle BAC = \angle DAC \rightarrow 2x+8=4x-12 \rightarrow 20=2x \rightarrow x=10$.

5 Kite — Main diagonal bisects cross diagonal. Find x:

$$BE = 3x - 1, \quad DE = 4x - 7$$

$$= x = 6, \quad BE = DE = 17$$

Main diagonal bisects BD: $BE=DE \rightarrow 3x-1=4x-7 \rightarrow 6=x$.

6 Kite — Vertex angle bisected at C. Find x:

$$\angle BCA = (x + 12)^\circ, \quad \angle DCA = (3x - 16)^\circ$$

$$= x = 14, \quad m\angle BCA = m\angle DCA = 26^\circ$$

Main diagonal bisects $\angle C$: $\angle BCA = \angle DCA \rightarrow x+12=3x-16 \rightarrow 28=2x \rightarrow x=14$.

7 Kite — Find x and $m\angle B$. Use sum of angles = 360° :

$$\angle B = \angle D = (3x + 5)^\circ, \quad \angle A = 80^\circ, \quad \angle C = 60^\circ$$

$$= x = 35, \quad m\angle B = m\angle D = 110^\circ$$

$\angle A + \angle C + 2\angle B = 360^\circ$: $80+60+2(3x+5)=360 \rightarrow 6x+150=360 \rightarrow 6x=210 \rightarrow x=35$.

8 Kite — Find x and the perimeter:

$$AB = AD = 4x - 1, \quad CB = CD = 2x + 3, \quad P = 40$$

$$= x = 3, \quad AB = 11, \quad CB = 9$$

$2(4x-1)+2(2x+3)=40 \rightarrow 12x+4=40 \rightarrow 12x=36 \rightarrow x=3$.

